

# Session 1

## Computing, Robotics and Bionics

**Sergio Sama Rubio**

Sant Cugat del Vallès, 7-Sep-2026

# Course Materials

GENERAL BIOENGINEERING ...

Portal profesores

Students portal

¿Necesitas ayuda u orientación...

Avisos

Forum links

Forum para compartir links meet

Course notes

Grades

Final Grades

CRB1 2026-2027

Course schedule

Session 1

Computing, Robotics and Bionics 1 (2026/2027 1ST\_SEM)

Course Settings Participants Grades Activities More Open LMS

GENERAL BIOENGINEERING SPACE

Portal profesores

Hidden from students

Dejar oculto para estudiantes. Es accesible para profesores aunque esté en oculto.

Students portal

¿Necesitas ayuda u orientación sobre el entorno Moodle?

Si eres estudiante

# Introductions

Sergio Sama Rubio

<https://www.linkedin.com/in/sergisama/>

## Education

- Chemical Engineer (IQS, Ramon Llull University)
- Industrial Engineer (Ramon Llull University) PDD IESE

## Professional experience

- Baker Hughes, General Electric, Halliburton, AspenTech



# Class Structure

1. Regular: Slides + Python notebooks (in front of computer)
2. Miniprojects: teams of 2-3
3. Miniproject presentation: teams present the results of their Miniproject to two types of audiences
4. Checkpoints/Exam

In all cases:

- Bring computer
- Use of LLM: encouraged in general, except Theory portion of Checkpoints/Exam
- Language: strongly encouraged to use English for everything (particularly coding!)
- Work in teams
- Student Reps.?

# Course Schedule

# Pre-Requisites

- Google Chrome browser
- Google Account
- Google Drive + Google Drive Desktop application
- Github Account
- Google Colab (<https://colab.google/> , <https://colab.research.google.com/> )
- Kahoot

## Other

- ChatGPT Account
- Kaggle Account
- NCBI account (<https://account.ncbi.nlm.nih.gov/>) and API (<https://account.ncbi.nlm.nih.gov/settings/> and then Create API key)
- FDA API (<https://open.fda.gov/apis/authentication/> )

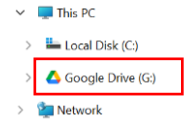
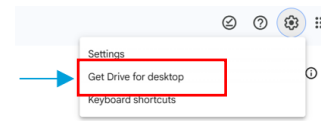
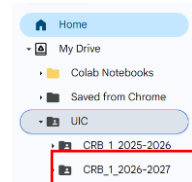
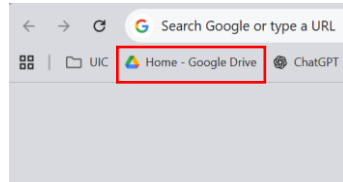
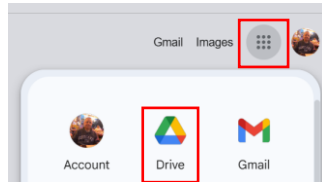
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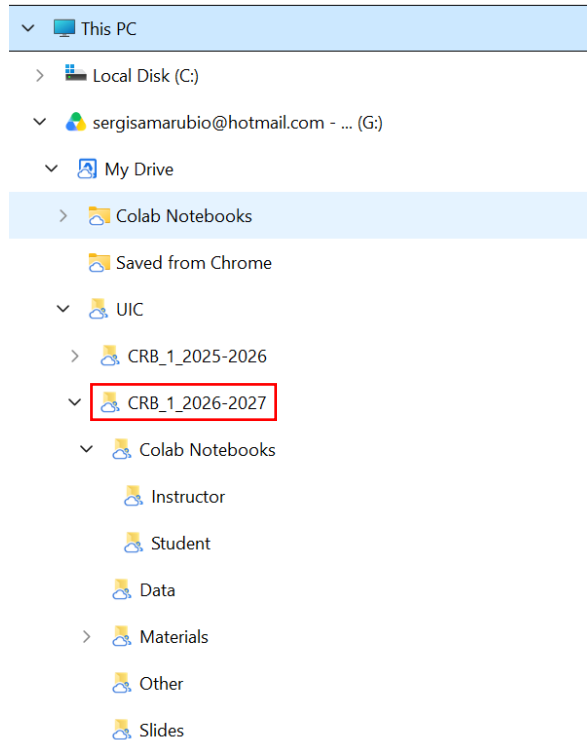
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<https://drive.google.com/drive/home>

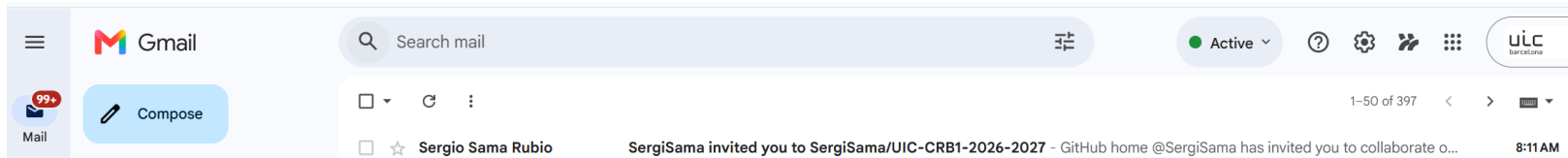


# Folder structure



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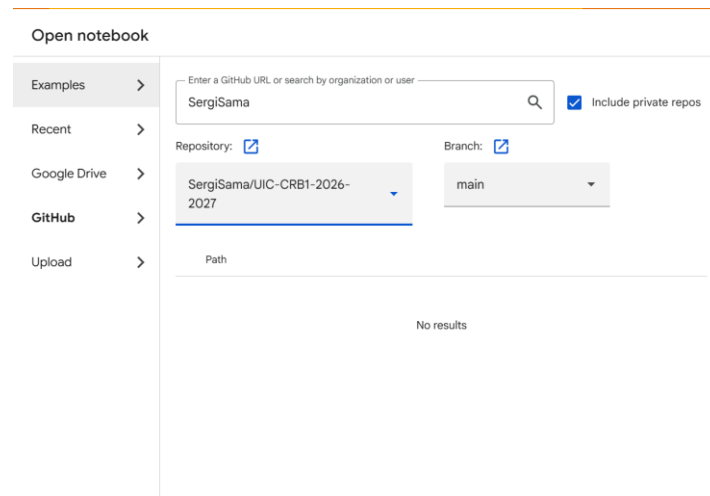
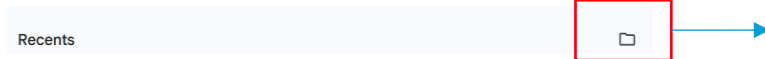
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# Google Colab

<https://colab.google/>

## Google Colaboratory

Colab is a hosted Jupyter Notebook service that requires no setup to use and provides free access to computing resources, including GPUs and TPUs. Colab is especially well suited to machine learning, data science, and education.



Cancel

# Start your engines!

<https://github.com/SergiSama/UIC-CRB1-2026-2027>

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# Q & A



# End of Session 1

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